Global Industrial Safety Initiative & Formal Sign-off Baseline Specification

Endogenous Inherently Secure Microarchitecture & Full-Domain Impedance Convergence Engineering Specification for Mission-Critical Heterogeneous SoCs (v2.0.0)

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Abstract

As mission-critical heterogeneous System-on-Chips (SoCs) deeply penetrate high-voltage automotive electric powertrains, autonomous driving computing platforms, smart electrical grids, heavy industrial automation, aerospace, and critical national infrastructure, modern semiconductor architectures are tightly integrating gigahertz-tier general-purpose processing cores, domain-specific accelerators, and high-voltage execution modules onto a single silicon substrate. Under such hostile operating conditions, ultra-fast switching transients (dv/dt) from wide-bandgap (SiC/GaN) power stages, high-altitude electromagnetic pulses (HEMP), single-event upsets (SEU), microarchitectural side-channel probing, instruction flow tampering, and unauthorized kernel memory dumping have emerged as persistent baseline operational threats.

Conventional rigid defense mechanisms—relying on supervisory software patches, external hard threshold clamping, brute-force reset circuits, and static cryptographic isolation—have definitively reached their fundamental physical and logical boundaries:

1. Abrupt current interruption under inductive loading induces catastrophic inductive kickback spikes ($V_{\text{transient}} = -L \cdot \frac{di}{dt}$), causing dielectric breakdown in sub-micron gate oxides and triggering regenerative parasitic silicon-controlled rectifier (SCR) latch-up;
2. Asynchronous clock-domain crossing (CDC) topologies undergo severe phase collapse under abrupt hard resets, stranding dirty “ghost states” across deep pipeline registers and asynchronous FIFOs, inducing unrecoverable bus protocol deadlocks;
3. Conventional processor execution pipelines remain physically and logically transparent, exposing speculative execution side-channels to dynamic instruction injection and covert telemetry exfiltration;
4. Static persistence of plaintext cryptographic keys within non-volatile media offers zero defense against micro-probing, focused ion beam (FIB) attacks, and differential/correlation power analysis (DPA/CPA).

This specification establishes the methodological foundation of Endogenous Impedance Convergence (EIC) and formalizes rigorous industrial sign-off criteria (v2.0.0 baseline) across three defensive layers: Logical Immunity, Physical Immunity, and Sovereignty Anchoring. By integrating on-die dynamic impedance-sinking topologies, out-of-band (OOB) hardware observers, dynamic instruction stream obfuscation, active white-noise side-channel flattening, and multi-stress drift-adaptive physically unclonable functions (PUF), the EIC architecture attenuates high-entropy disturbances within nanosecond regimes and achieves deterministic self-healing via pure hardware finite state machines within microsecond bounds.

This specification establishes six mandatory sign-off criteria:

1. Temporal coverage of parameterized SystemVerilog Assertions (SVA) must be formally exhaustive, enforcing a formal mutation score exceeding 95%;
2. Hardware instruction authentication over discrete clock cycles ($T_0 \sim T_n$) must achieve 100% temporal coverage, with unauthorized probes triggering bounded compute collapse;
3. Hardware-enforced interception of unauthorized kernel dumps must redirect bus transactions to a high-fidelity decoy sandbox within an isolation latency $T_{\text{isolate}} \leq 4$ clock cycles;
4. Hardware-grade out-of-band (OOB) isolation slices must retain autonomous, bottom-up safe-reload sovereignty independent of software intervention;
5. Dynamic side-channel power flattening and anti-magnetic write-through topologies must suppress external leakage entropy below mathematically provable bounds;
6. A zero-plaintext Root-of-Trust engine anchored on dynamic PUF architectures must guarantee error-free key reconstruction across the full temperature envelope (-40°C to +150°C) and multi-stress operational lifespans.

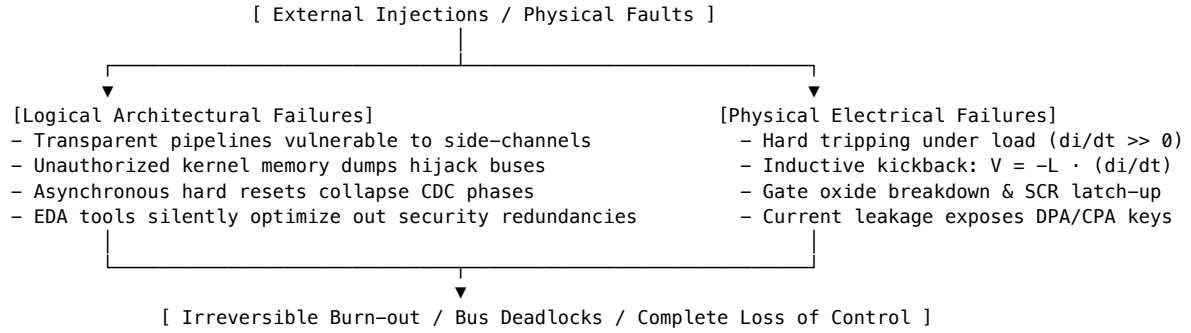
This specification defines unified industrial evaluation interfaces and black-box verification metrics; no proprietary Register-Transfer Level (RTL) source code or silicon layout topologies are disclosed.

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Chapter 1: Physical Limits and System-Induced Failures of Rigid Defenses

Under topologies where heterogeneous compute clusters tightly couple with high-power physical actuators, rigid defenses relying on software watchdogs, passive filtering, and hard tripping exhibit five critical physical and architectural failure modes:



1.1 Instruction Stream Transparency and Unauthorized Kernel Dump Hijacking

Traditional processor pipelines assume an execution environment of implicit trust. Attackers injecting microarchitectural glitches (voltage/clock fluctuations) or exploiting speculative execution side-channels routinely bypass operating system virtual memory protections:

- **Instruction Tampering & Injection**: Malicious instructions are fetched and executed by out-of-order dispatch units due to the lack of cycle-accurate hardware authenticity checks;
- **Unauthorized Kernel Dumps**: Upon encountering critical assertion faults, conventional trap routines dump raw register and memory contexts onto unprotected interconnects, exposing kernel-privileged memory footprints to external logical analyzers.

1.2 The Inductive Overshoot Paradox and Physical Destruction

In electric drivetrains, industrial motor controls, and grid inverters, power switches drive highly inductive physical plants. When overcurrent faults are detected, rigid protection circuits abruptly pull gate voltages low to achieve microsecond-scale trip times:

$$V_{\text{transient}} = -L \cdot \frac{di}{dt}$$

Because inductor currents cannot change discontinuously, forced cut-off drives $\frac{di}{dt} \to \infty$, triggering transient inductive kickback spikes of several kilovolts. This overvoltage breaches the thin gate oxide of sub-micron logic chips long before external transient voltage suppressors clamp the rail. Furthermore, it triggers regenerative latch-up across parasitic substrate NPN-PNP structures, culminating in secondary breakdown, thermal runaway, and permanent hardware burn-out.

1.3 Physical Side-Channel Power Leakage and Electromagnetic Vulnerability

When cryptographic primitives execute within static digital logic, CMOS switching currents correlate strongly with the Hamming weight and Hamming distance of the processed operands. Without breaching package encapsulation, adversaries utilizing high-bandwidth current probes or near-field EM antennae can reconstruct 128-bit or 256-bit cryptographic keys within thousands of clock cycles via Differential Power Analysis (DPA) or Correlation Power Analysis (CPA).

1.4 Static Root-of-Trust Solidification and Environmental Aging Degradation

Traditional hardware Roots of Trust (RoT) rely heavily on burning static plaintext keys into on-chip eFuses, mask ROMs, or external non-volatile memories:

- Static storage offers zero physical defense against focused ion beam (FIB) probing or backside laser voltage scanning;

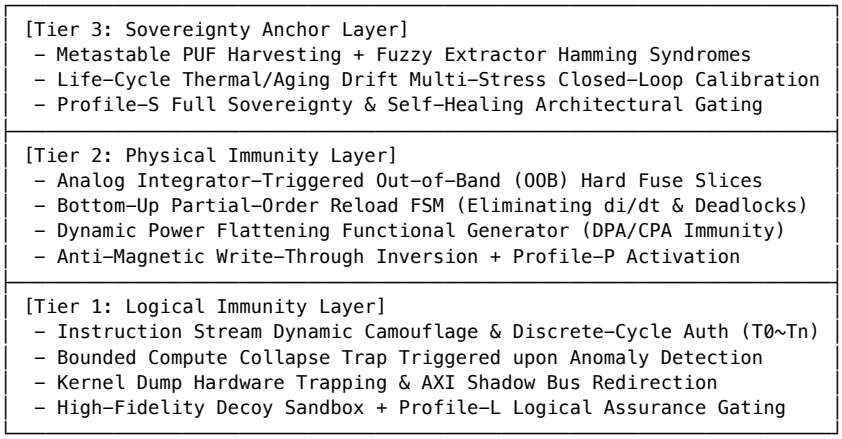
- Early silicon PUFs extract raw entropy without multi-stress environmental adaptation. Susceptible to temperature gradients (-40°C to +150°C), supply noise, and Negative Bias Temperature Instability (NBTI) aging, uncompensated PUFs experience surging Bit Error Rates (BER) over prolonged deployments, causing cryptographic authentication collapse.

1.5 Clock-Domain Crossing (CDC) Phase Collapse and EDA Synthesis Traps

- **CDC Ghost States**: Applying uncoordinated asynchronous resets across asynchronous boundaries drives multi-stage synchronizers into metastability. Residual dirty states trapped within deep pipeline stages and cross-domain FIFOs induce permanent bus protocol deadlocks upon reboot;
- **EDA Synthesis Traps**: Commercial logic synthesis and place-and-route (P&R) suites optimize relentlessly for Power, Performance, and Area (PPA). Consequently, dual-rail state machines, shadow registers, and de-glitch bridges are routinely classified as "redundant logic without functional load" and silently pruned, leaving silicon physically defenseless.

Chapter 2: Endogenous Impedance Convergence (EIC) Full-Stack Microarchitecture

Abandoning passive interception and rigid tripping, the Endogenous Control paradigm establishes three concentric tiers of autonomous silicon immunity:



2.1 Logical Immunity Layer

Operating across execution pipelines and master interconnect fabrics, this layer immunizes the instruction flow:

- **Dynamic Logic Camouflage & Instruction Authentication (EIC-SPEC-01)**: The execution pipeline integrates discrete-beat hardware authenticators tied to temporal causal chains ($T_0 \sim T_n$). Instructions undergo dynamic hardware decamouflaging at the fetch-decode boundary; instructions failing state-machine verification are denied valid execution contexts. Upon consecutive invalid operations, gate-level compute collapse engages, forcing execution pipelines into deterministic dead-loop traps to halt malicious payloads.
- **Cross-Layer Exception Trapping & Decoy Sandboxing (EIC-SPEC-02)**: Hardware-level transparent monitors at AXI/AHB bridge boundaries audit kernel invocation manifolds against formal transaction invariants. Unauthorized memory dumps are intercepted prior to interconnect egress. Concurrently, zero-latency shadow bus multiplexers redirect malicious probes to an on-chip, high-fidelity Decoy Sandbox. The sandbox streams structurally compliant, entropy-scrambled telemetry, binding attackers to synthetic state spaces for non-intrusive threat attribution.

2.2 Physical Immunity Layer

Operating at the physical boundaries and analog energy interfaces, this layer guarantees continuous disturbance dissipation:

- **Out-of-Band Analog Hard Fuse & Partial-Order Reload (EIC-SPEC-03)**: An autonomous Out-of-Band (OOB) hardware slice operates electrically isolated from general compute clusters. Pure analog continuous-time integrators resolve high-voltage transients (dv/dt), superseding software routines or digital comparators prone to state corruption. Under severe overshoots, on-chip dynamic impedance surfaces reshape the transient profile, steering inductive energy into low-entropy dissipative sinks to keep transient potentials below dielectric breakdown thresholds. Following fault clearing, an autonomous hardware sequencer executes a strict partial-order reset handshake, flushing asynchronous FIFOs, clearing CDC synchronizers, and restoring link states to permanently eliminate ghost deadlocks.
- **Side-Channel Physical Immunity & Power Flattening (EIC-SPEC-04)**: Real-time complementary current-injection circuits introduce anti-phase stress currents and Gaussian white noise during cryptographic executions. By flattening the total current consumption profile, the externally observable power trace decouples completely from internal Hamming weights, eliminating the mathematical basis of DPA/CPA attacks. Concurrently, an orthogonal phase-inversion layout suppresses eddy currents induced by external electromagnetic pulses (EMP), preventing bit-flips across sensitive nodes.

2.3 Sovereignty Anchor Layer

Establishing the cryptographic foundation of silicon integrity and capability licensing:

- **Physically Unclonable Feature Anchoring & Dynamic Calibration (EIC-SPEC-05)**: Entropy harvesters are located within isolated cleanroom silicon domains with zero direct electrical coupling to the system interconnect. Intrinsic sub-micron process variations are digitized into unique spatial fingerprints. Dedicated hardware fuzzy extractors apply BCH/Hamming error-correction polynomials to reconstruct invariant cryptographic keys without plaintext exposure.
- **Multi-Stress Adaptive Closed-Loop Calibration**: On-die environmental sensor arrays integrate thermal gradients (-40°C to +150°C) and device aging degradation (NBTI/HCI) over operational lifetimes, continuously updating the algebraic decoding parameters of the fuzzy extractor to ensure zero-BER key reconstruction throughout the silicon lifecycle.
- **Zero-Plaintext Persistence & Architectural Profile Gating**: No plaintext cryptographic keys persist across non-volatile memory arrays. Ephemeral root keys are derived on-demand, utilized within single execution phases, and wiped via hardware zeroization. Paired with cryptographic signatures, this engine controls pure-hardware architectural feature-flag gates, enabling standardized execution profiles (Profile-L, Profile-P, Profile-S).

Chapter 3: Formal Sign-off Hard Gate Matrix (v2.0.0 Baseline)

To supersede empirical assertion monitoring and unconstrained Constrained Random Verification (CRV), all silicon implementations claiming EIC compliance must formally satisfy the following mathematical sign-off invariants prior to tape-out:

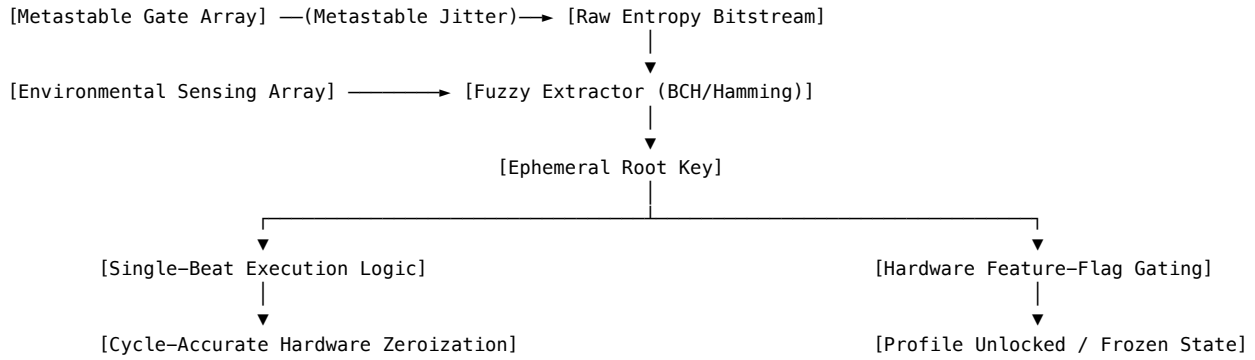
Rule ID	Gate Definition	Invariant Specification	Severity	Coverage
SVA-LINT-01	Unbounded Delay Blocking	Forbid unbounded temporal operator `##[0:$]`; enforce finite upper bound $k \leq N_{\text{cycles}}$ to prevent state explosion.	Blocking	Universal Baseline
SVA-LINT-02	Reset Contract Enforcement	All monitoring properties must explicitly bind to active-low reset trees: `disable iff (!rst_n)`.	Blocking	Universal Baseline
SVA-LINT-03	Preponed Phase Sampling	Mandate variable value sampling via `$sampled` in preponed clock phases to eliminate simulation race conditions.	Blocking	Universal Baseline
SIG-GATE-01	Global Deadlock Elimination	Model checkers must exhaustively prove the absence of circular wait states or deadlocks across all state-machine transitions.	Sign-off Gate	EIC-SPEC-02 / 03
SIG-GATE-02	Cross-Coverage Threshold	Achieve 100% cross-coverage across defensive assertion meshes, FSM state-transition matrices, and AXI handshakes.	Sign-off Gate	Universal Baseline
SIG-GATE-03	Formal Mutation Score	Formal mutation score under fault injection must exceed 95%, eliminating false-passes and dead logic.	Sign-off Gate	Universal Baseline
SIG-GATE-04	Bounded Instruction Auth	Hardware authenticators must trap illegal instruction injections and initiate compute collapse within $T_{\text{auth}} \leq 2$ cycles.	Sign-off Gate	EIC-SPEC-01
SIG-GATE-05	Decoy Isolation Contract	Shadow bus redirection to decoy sandboxes must complete within $T_{\text{isolate}} \leq 4$ cycles without master bus data residue.	Sign-off Gate	EIC-SPEC-02
SIG-GATE-06	Bounded OOB Safe Reload	System must drain deep asynchronous FIFOs and re-establish baseline communication via partial-order resets within bounded cycles.	Sign-off Gate	EIC-SPEC-03
SIG-GATE-07	Side-Channel Entropy Bound	Cross-correlation between complementary current injections and internal operations must verify $R_{\text{corr}} < \epsilon_{\text{threshold}}$.	Sign-off Gate	EIC-SPEC-04
SIG-GATE-08	PUF Drift Calibration	Fuzzy extractor polynomials must guarantee key reconstruction failure probability $P_{\text{fail}} < 10^{-9}$ across full thermal and aging spans.	Sign-off Gate	EIC-SPEC-05

Chapter 4: Hardware Root-of-Trust, Stress Compensation & Architectural Profiles

4.1 Out-of-Band Physical Cleanroom Domain Isolation

- Entropy-harvesting circuits reside within a physically segregated Cleanroom Domain on the die floorplan;
- The cleanroom domain is powered by a dedicated low-noise analog Low-Dropout (LDO) regulator island, electrically severing substrate noise coupled from high-frequency digital logic;
- Only unidirectional, cryptographically authenticated handshakes interface the cleanroom with the system interconnect; unconditioned analog entropy signals are strictly barred from the system Network-on-Chip (NoC).

4.2 Runtime Derivation and Zero-Plaintext Persistence



- Upon power-up or dynamic re-authentication, the PUF array engages ephemerally, prompting the fuzzy extractor to synthesize the deterministic root key using real-time drift compensation matrices;
- Following cryptographic validation, dedicated hardware state machines enforce cycle-accurate zeroization across all local holding registers, eliminating remanence and cold-boot vulnerability vectors.

4.3 Architectural Capability Profiles and Hardware Gating

Through cryptographic gating enforced by EIC-SPEC-05 and EIC-SPEC-01, EIC black-box IPs expose modular architectural profiles:

- **Profile-L (Logical Assurance Profile)**: Engages EIC-SPEC-01 (Dynamic Instruction Camouflage) and EIC-SPEC-02 (Decoy Sandboxing), securing instruction pipelines and bus transactions against tampering without requiring analog silicon extensions.
- **Profile-P (Physical Resilience Profile)**: Augments Profile-L with EIC-SPEC-03 (Analog OOB Fuse & Partial-Order Reset) and EIC-SPEC-04 (Power Flattening & Anti-Magnetic Topologies), conferring robust immunity against inductive overshoots (dv/dt) and side-channel power extraction in automotive and industrial drives.
- **Profile-S (Sovereign Root Profile)**: Enables the full defensive suite (EIC-SPEC-01 through EIC-SPEC-05), activating multi-stress drift-compensated PUF anchors to establish a closed-loop “Sensing-Dissipation-Defense-Healing” sovereign silicon foundation.

Chapter 5: Zero-Knowledge Evaluation Interface & Formal Sign-off Protocol

To facilitate the cross-enterprise adoption of EIC microarchitectures, Jianyuan Lab defines an asymmetric black-box evaluation framework. Through its exclusive commercial distributor (Hangzhou Yuemu Network Technology Co., Ltd.), verified semiconductor design houses, automotive Tier-1 functional safety units, and formal verification laboratories can perform sign-off audits under complete intellectual property isolation.

5.1 Asymmetric Black-Box Verification & Zero-RTL Doctrine

- **Standardized Behavioral Models**: Evaluators receive an air-gapped, pre-compiled behavioral shared library (`.so`), parameterized AXI4 formal stub wrappers, and standardized SVA mutation suites within the `JYL-EIC-SDK-v2.0.0-Evaluation` package;
- **Absolute Asset Isolation**: Delivery packages utilize one-way algebraic projection across control and data paths, strictly excluding synthesizable Register-Transfer Level (RTL) code, gate netlists, or GDSII layout geometries;
- **Bidirectional Isolation**: All evaluation executions run locally within the licensee's air-gapped test harnesses. The licensor neither requests nor inspects the licensee's proprietary codebase or functional vectors.

5.2 Normalized Trace Interchange Specification

- The sole medium of technical alignment is strictly limited to strongly typed, normalized execution trace logs adhering to `ipc_trace_schema.json` and formal mutation coverage reports;
- Technical sign-off evaluates whether state-transition matrices converge, trapping latencies remain bounded ($T_{\text{auth}} \leq 2$), and decoy redirection completes within $T_{\text{isolate}} \leq 4$, eliminating subjective interpretations.

5.3 Active Defensive Modeling and Cryptographic Time-Lock Mechanisms

Evaluation packages incorporate active defensive barriers to preserve evaluation integrity:

- **Control-Flow Watermarking & Anti-Debugging**: Compiled binaries utilize control-flow flattening, entangling licensee identities with dispatch branch tables while enforcing anti-ptrace and anti-memory-dumping guards;
- **Parallel Decoy State Engines**: Non-standard probing or clock glitching triggers automated diversion to a synthetic Decoy Sandbox, returning protocol-compliant Gaussian-flattened telemetry to invalidate unauthorized reverse-engineering;
- **10-Day Cryptographic Time-Lock**: Licenses strictly enforce a 10-calendar-day evaluation window (fault-tolerant extension up to 4 days; total lifetime capped at 14 days). Epoche timestamps seed internal round-key derivations; expiration collapses internal FSM algebraic bases, defeating binary patching;
- **Forensic Canary Tokens**: Invariant algebraic manifolds are embedded within the behavioral logic. Unauthorized silicon tape-outs incorporating stolen structural primitives will irrevocably reproduce this algebraic signature, serving as prima facie evidence of trade secret misappropriation.

5.4 Controlled Evaluation Access Protocol

- **Entity Pre-Qualification**: Access is restricted to established entities demonstrating verifiable semiconductor track records, automotive powertrain competencies, or industrial PLC verification infrastructure;
- **Contractual Safeguards**: Inquiries originating from corporate domains must be addressed to `yuemu.sec@proton.me`. Qualified applicants must execute a bilaterally binding EULA enforcing a 5-person full-time personnel whitelist, cleanroom development isolation, strict prohibition of AI/LLM code ingestion, and a grant-back covenant on derivative architectures;
- **Unidirectional Secure Delivery**: Upon legal verification, bespoke behavioral binaries embedding unique forensic canary seeds are compiled offline and securely delivered.

Chapter 6: Call to Action for the Global Semiconductor Industry

1. **To International Functional Safety Standards Bodies (ISO 26262 / IEC 61508 / IEC 62443 / DO-254)**: Formally incorporate "Formal Mutation Coverage (Mutation Score > 95%)," "Non-Intrusive Hardware Decoy Sandboxing," and "Autonomous Out-of-Band Physical Self-Healing" into mandatory sign-off criteria for next-generation safety-critical silicon certifications.
2. **To Fabless Chip Designers and System Architects**: Relinquish the brittle assumption that lower-tier silicon vulnerabilities can be perpetually mitigated via software patches. Allocate non-negotiable silicon floorplan area for endogenous impedance sinks, hardware instruction-flow authenticators, and autonomous reset FSMs at the architectural inception phase.
3. **To Commercial EDA and Formal Verification Vendors**: Establish architectural hardening primitives within synthesis and place-and-route toolchains. Provide native "Don't Touch" isolation for dual-rail state machines, shadow interconnect logic, and formal contract checkers, preventing automated optimization engines from silently pruning mission-critical security logic.
4. **To Mission-Critical System Integrators and Automotive Tier-1s**: Audit semiconductor supply chains for physical-boundary failure modes. Mandate that semiconductor vendors furnish EIC v2.0.0 formal sign-off certificates, verification logs, and out-of-band self-healing demonstrations to insulate mission-critical platforms against catastrophic field disruptions.

Archival & Statutory Attribution

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  1. CNIPA Application No. `202611592366.1` (Docket: Z262135TPCN): *A Dynamic Logic Camouflage and Instruction Flow Authentication System and Method for Black-Box Environments*
  2. CNIPA Application No. `202611592386.9` (Docket: Z262136TPCN): *A Cross-Layer Anomaly Monitoring and Non-Intrusive Decoy Isolation System and Method*
  3. CNIPA Application No. `202611592364.2` (Docket: Z262137TPCN): *A Heterogeneous Out-of-Band Physical Hard-Fuse and Bottom-Up Safe-Reload System and Method Resistant to Common-Cause Failures*
  4. CNIPA Application No. `202611592385.4` (Docket: Z262138TPCN): *A Side-Channel Physical Immunity Architecture and Method Based on Anti-Magnetic Inversion and Active White-Noise Power Flattening*
  5. CNIPA Application No. `202611592365.7` (Docket: Z262139TPCN): *A Hardware Intellectual Property Anchoring and Dynamic Calibration System and Method Based on Physically Unclonable Features*

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